Checkpoint I — Study Plan and Tracking Setup

Title

Play & Learn: Quantum Algorithms

Description

I aim to dive into rigorous quantum algorithm material. All research papers and artistic projects are built upon quantum algorithms, though I used to skip algorithm sections in papers as they were difficult to follow. In this course, I will collect and study quantum algorithms, implement them on Gemini Lab, improve my workflow, and expand my toolkit. If possible, I hope to develop new quantum algorithms.

Objectives

1. Make a list of Quantum Algorithms, develop a deeper understanding of them various algorithms and how they can be applied within musical practice.
2. Attempt to replicate these algorithms on Gemini Lab and QAC Toolkit.
3. produce an (maybe not complex) algorithm as the basis for my artworks or papers.
4. Optional: Artistic-related Quantum Algorithms learning path.

Final deliverable

1. GitHub repositories with Codebase, Dataset, and Application. Specific content include below:
    - Quantum Algorithms study notes
    - Experiment file and data recorded
    - Artistic practice project
    - Journals, Documents, Assets
2. A presentation on slides at the end of semester, share to all xQAP members.

Week-by-week plan

Note: Since it is hard to predict the workload before starting, I use a syllabus structured by phases and weeks for quantification.

Phase I Attempts based on the Gemini Lab handbook and its built-in experiments. Week 3-8

- What are Quantum Algorithms?
  - Their role within quantum computing
  - Their role in artistic practice
  - Approaches to extracting and defining the quantum algorithms deployed within a performance or research project.
- Crate a list of quantum algorithms
  - strategies for classification
  - frameworks to quantify learning difficulty.
  - decide the study subsequence
- First quantum algorithm: starting from Gemini Lab experimentations (1-2 weeks), A complete learning cycle should include：
  - Quantum Algorithms study notes
  - Experiment file and record
  - Artistic practice project

    - Journals, Documents, Assets
- More quantum algorithm on handbook and machine.

Midterm

- Design a simple quantum algorithm and execute it on Gemini Lab.

Phase II Replicate existing quantum computing and art practice algorithms. Week 9-13

- After hands-on experience with hardware-oriented algorithms, we can move to algorithmic in papers/conferences/performance. we will not only replicate the algorithms, but also recreate their application and secondary creation within musical practice.
    - Integrate the QAC toolkits at this stage.
    - Explore workflows for reproducing these algorithms for artistic practice.
    - Note that certain algorithms may not be readily implementable within och.microqiskit, but experimental attempts are still good.
- More quantum algorithms.

Final Deliverable

- Develop a custom quantum algorithm
    - This may evolve from the midterm prototype or from other sources.
    - Embed this algorithm into a project, paper, or just presentation.
- Recap all path with a presentation on slide.

Milestones

- Every project after recreated
- 2 custom quantum algorithm
- Documented Artistic-related Quantum Algorithms learning path

Bibliography and resources

1. SPINQ Gemini Lab Quantum Computing Experimental Handbook
   This is a good starting point for learning both the algorithms and the device itself.
2. ISQCMC
   This is an excellent conference that brings together many algorithms. I can select some from papers and performances for my study.
3. Old qiskit text book
4. Quantum Computing for Everyone
5. Quantum Computing Algorithms: Discover how a little math goes a long way

Estimated workload

6 hours per week

tracking system

Link: https://hlx9751.github.io/project-tracker

A quantifiable and flexible tracking system built according to the progress of each project.

GitHub Repo: https://github.com/hlx9751/Play-Learn-Quantum-Algorithms

AI-use disclosure

I build the tracking system’s webpage and check my English writing with AI